

Real-Time Structural Topology Optimization

A Deep Learning Pipeline Bypassing Iterative FEA
via Convolutional Neural Networks

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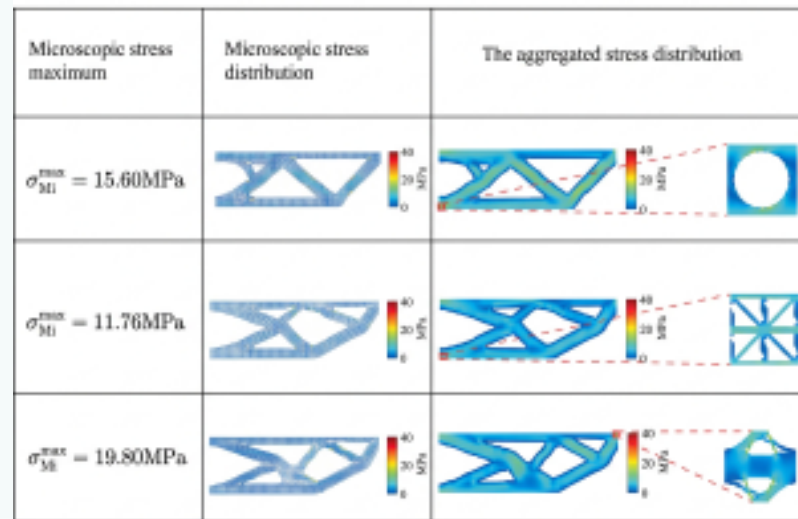
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The Research Problem

- **The Computational Bottleneck:** Traditional Topology Optimization (TO) relies on Finite Element Analysis (FEA) to find optimal material distributions.
- **Iterative Exhaustion:** Reaching a converged mathematical layout typically requires 80–150 sequential numerical solver iterations.
- **The Objective:** To formulate topology optimization as a data-driven, image-to-image regression task.
- **The Goal:** Predict global equilibrium structural layouts in a single forward pass without sacrificing physical validity.



Phase 1: Dataset Generation

SIMP Solver Implementation

We will implement a vectorized 2D SIMP (Solid Isotropic Material with Penalization) algorithm to act as our mathematical ground truth generator.

A mesh-independency sensitivity filter was applied to prevent checkerboard instabilities across the defined 60×120 element grid design domain.

Data Distribution

A multi-core generation pipeline was utilized to produce a robust dataset of 2,000 paired training samples.

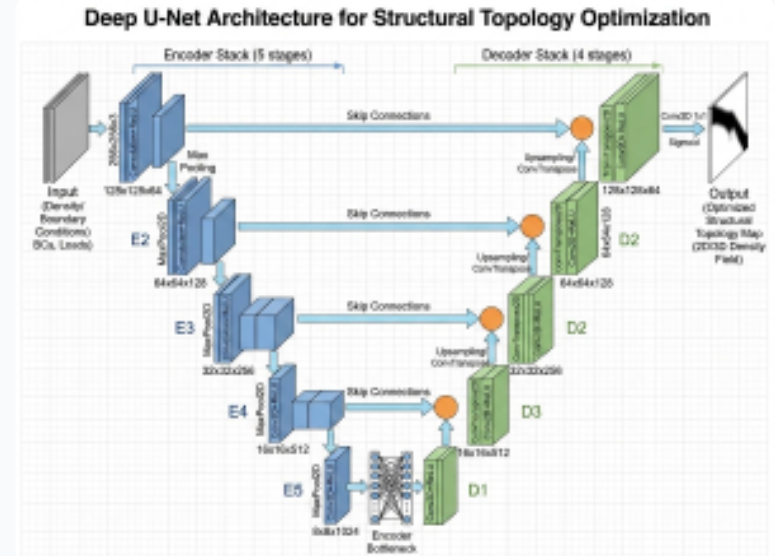
- **Variables:** Target volume fractions (V_f) were randomized uniformly between 0.2 and 0.6.
- **Load Cases:** Randomized 1–3 distinct nodal loads applied at varying angles and magnitudes on boundary edges.

Phase 2: Model Architecture

Input Encoding: The input is structured as a 5-channel spatial tensor defining boundary conditions, load vectors (F_x , F_y), and the target volume fraction.

Deep U-Net Selection: U-Net CNN architecture consisting of a 5-stage encoder stack and a 4-stage decoder stack.

Skip Connections: Essential for mapping both the global context (force paths) and local precision (strut intersections), allowing the model to bypass intermediate "gray" densities.



Phase 3: Training Strategy & Loss

To ensure physical realism rather than just pixel-accuracy, a multi-objective composite loss function was engineered.



Reconstruction Loss

Binary cross-entropy will be used to measure the direct deviation between the predicted density field and the exact SIMP-generated ground truth.



Volume Constraint

A strict penalty will be added to penalize deviations from the requested material volume fraction, ensuring the model obeys resource budgets.



Binarization Penalty

An aggressive mathematical penalty suppressing intermediate "gray" density predictions, driving network outputs toward definitive solid (1) or void (0) structures.

Phase 4: Testing & Validation

Target to achieve

< 30

Milliseconds Inference Time

Physical Feasibility Achieved

The model will be tested against 500 unseen validation geometries, proving its ability to generalize.

- **Acceleration:** To achieve a good speedup compared to traditional numerical iterations.
- **Compliance Check:** Then Predicted structures will be fed back into an FEA environment to verify relative error.

Project Progress Overview

Phase 1 is officially completed. Moving into model architecture, training, and validation pipeline.

Code is
ready

Phase 1

Dataset Generation

- SIMP Solver
- 3,000 Samples
- Code is ready, need to run code and generate dataset.

IN PROGRESS

Phase 2

Model Architecture

- Deep U-Net
- 5-Channel Input
- Skip Connections

UPCOMING

Phase 3

Training & Loss

- Reconstruction
- Volume Penalty
- Binarization

UPCOMING

Phase 4

Testing & Validation

- FEA Check
- < 30ms Speed
- Generalization